

Sustainable Approach to Fabricate High-Performance Symmetry Supercapacitor Electrodes from Natural Coconut-Shell-Derived Porous Activated Carbon with Nickel Oxide Nanocomposites

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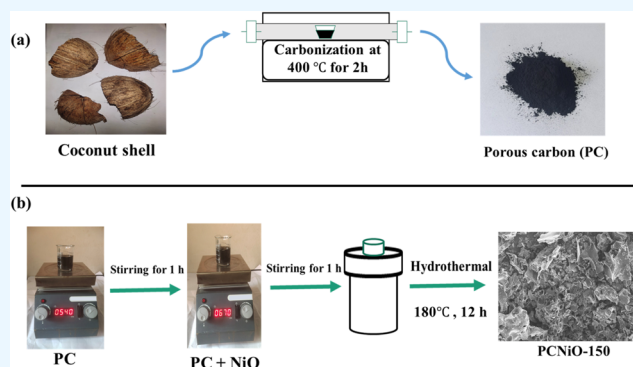
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ABSTRACT: This paper reports high specific capacitance of an activated carbon nickel oxide nanocomposite (PCNiO) electrode that has been synthesized from natural coconut shell using carbonization and an activated PCNiO nanocomposite with the help of a hydrothermal process. The structural phase, chemical change, morphology, and pore structure of the PCNiO nanocomposite were investigated using a variety of techniques including X-ray diffraction (XRD), Fourier transform infrared (FTIR), Brunauer–Emmett–Teller (BET), thermo-gravimetric analysis (TGA), Raman spectroscopy, field emission scanning electron microscopy (FESEM), and high-resolution transmission electron microscopy (HRTEM) techniques. Among the prepared samples, PCNiO-150 displays the most significant characteristics that were used to create symmetric supercapacitors (SSCs). It had a specific capacitance (C_{sp}) of 598.6 F/g at a scan rate of 10 mV/s. The Galvanostatic charging–discharging (GCD) curves showed a high specific capacitance (C_{sp}) of 656.2 F/g at a current density (CD) of 1.5 A/g. Additionally, even after 5000 cycles, it had achieved long-term cycle stability with capacitance retention of 78.34% and Coulombic efficiency of 97.55%. Its highest energy density (ED) and power density (PD) were 44 Wh kg^{−1} and 562.5 W kg^{−1}, respectively. Additionally, the fabricated SSC device is serially connected to turn on a commercial green LED for 30–40 s at the time of the experiment. This paper proposes a novel environmentally sustainable and easy-to-use carbon source as well as a cost-effective and technologically unique approach for carbon supercapacitors in environmental applications.



1. INTRODUCTION

The development of sustainable and clean energy technology is a challenge for researchers and industries to fulfill demands and also prevent the global energy crisis and environmental degradation.^{1,2} However, high-efficiency sustainable energy generation and storage remain a serious issue despite improvements in advanced materials and technology for the applications of energy conversion, harvesting, and storage.³ Due to their exceptional efficiency in energy harvesting, supercapacitors and overall water splitting have recently gained popularity as emerging technologies for electrochemical energy storage and conversion.^{4–7} Additionally, supercapacitors have a number of desired characteristics, including a high power density, quick charging–discharging rates, and exceptional cycle stability.^{8–10} The supercapacitors are divided into two categories based on the charge storage method: (i) electric double-layer capacitors (EDLCs) and (ii) pseudocapacitors.^{11–13} The accumulation of electrolyte ions at the electrode and electrolyte contact in EDLCs drives a physical mechanism. The rapid surface redox reactions at the electrode–electrolyte

interface are the focus of the faradaic process, which serves as the framework for pseudocapacitors.^{14,15} EDLC-based capacitors have relatively low energy density, and the pseudocapacitance materials provide higher energy density but have lower power density. The proper synthesis and fabrication strategy of these two materials can be expected to significantly improve the performance of supercapacitors.¹⁶ The electrode material is a crucial component of a supercapacitor because it impacts the device's major performance.¹⁷ There are several factors involved, such as the chemical composition, electrical conductivity, specific surface area (SSA), and the structure of electrode materials, which are important factors affecting the

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electrochemical properties of supercapacitor electrodes.¹⁸ Recently, porous activated carbon-based materials have attracted greater attention due to their high specific surface area (SSA), superior electrical conductivity, and environmentally friendly, cost-effective, and large-scale production. Furthermore, porous carbon materials provide a large active site for electrochemical reactions, offer a short ion transportation diffusion pathway, and are highly recommended for high-performance electrode materials.¹⁹ In addition, the introduction of heteroatoms can significantly improve their electron-transfer capacity.²⁰

Recently, agricultural products have received considerable attention as carbon precursors for generating activated carbon (AC), mainly because of their renewable, low-cost, and environmentally friendly nature.²¹ We developed a supercapacitor electrode of activated carbon-based biomass renewable material, which offers a more efficient method for exploitation of renewable natural resources and is also an effective solution to the scarcity of fossil fuels.²² From literature reviews, a few are reported: activated carbon was synthesized from agricultural waste such as orange peel,²³ corn husk,²⁴ coconut shell,²⁵ eucalyptus globulus seed (EGS),²⁶ peanut shell,²⁷ banana stem,²⁸ and onion peel.²⁹ Among the various biomasses, coconut shells are an available agricultural waste product that can be used as a desirable raw material to produce highly valued products like activated carbon and graphene materials owing to their high carbon content and availability as a byproduct of the coconut industry.³⁰ The process of converting it into activated carbon improves its value and yields a sustainable and ecofriendly material for a range of uses, such as carbon-based electrodes. Activated carbon derived from coconut shells is preferred owing to its low ash, porous structure, high water solubility, and efficient pyrolysis method. Its small particle size improves the heating efficiency, resulting in enhanced porosity and a high surface area, making it suitable for supercapacitor applications.³¹ Additionally, it was claimed that activated coconut fiber carbon (ACFC) has excellent electrical and mechanical properties, is nontoxic, affordable, widely available, commercially viable technology, and with a high production rate.³²

In contrast to EDLC, pseudocapacitors (PCs) are subjected to redox reactions and tend to exhibit relatively high energy densities.³³ Metal oxides have been regarded as one of the most promising candidates due to their low cost, energy density, and environmental benefits.³⁴ For energy storage applications, several types of transition-metal oxides, including Mn_3O_4 , NiO , V_2O_3 , MoO_3 , Co_3O_4 , and CuO , have been studied as pseudocapacitive components in carbon-based electrodes.^{32,35–39} Among the transition-metal oxides, nickel oxide nanoparticles are considered as composite potential alternative electrode materials for pseudocapacitors due to their low-cost effectiveness, nonnoble nature, natural abundance, and high theoretical specific capacitance.⁴⁰ There are various techniques to fabricate activated biochar derived from agricultural waste incorporated with nickel oxide nanocomposites, such as hydrothermal synthesis,⁴¹ calcination at different temperatures,⁴² low-temperature water-bath method,⁴³ and carbonization process.⁴⁴ However, the long preparation time, complex processing methods, and high costs have limited their application. In this case, the development of an environmentally friendly method and low-cost carbon material is a critical issue. Currently, biomass-derived activated carbon materials have emerged as a type of

electrode material owing to their low price, accessibility, and multiple sources. However, few research studies have been conducted on the deposition of NiO nanoparticles onto activated carbon derived from biomass for supercapacitor applications. For instance, walnut shell-derived activated carbon altered with nickel oxide (NiO) nanoparticles exhibited an enhancement in the specific capacitance of the supercapacitor electrode by 16% in comparison to the original unmodified activated carbon.⁴⁵ Water hyacinth-derived activated carbon decorated with NiO has been prepared by a hydrothermal process, yielding a high specific capacitance of 249.5 F g^{-1} at 1 A g^{-1} .⁴⁶ Activated carbon (AC) obtained from coconut shell via chemical activation utilizing H_3PO_4 as the activating agent and incorporating NiO for the production of NiO/AC composites demonstrated a specific capacitance of 142 F g^{-1} at 6 mA g^{-1} .⁴⁷ Activated carbon materials from guava leaves were produced using a simple chemical activation method with KOH and dispersed NiO nanocrystals via hydrothermal processing, achieving a specific capacitance of 461 F g^{-1} at 2.3 A g^{-1} .⁴¹ Nevertheless, most of them do not show very high specific capacitance and energy density values. Additionally, the development of an environmentally friendly and low-cost method for incorporating NiO nanoparticles in porous carbon for supercapacitors is imperative, given the extensive commercial applications of electrode materials.

In this study, we described a cost-effective and environmentally friendly method for preparing activated porous carbon derived from coconut shells incorporated with nickel oxide nanocomposites through two-step carbonization followed by a hydrothermal process. Our work utilizes renewable coconut-shell resources in a novel way, contributing to the resolution of sustainability issues and minimizing the ecological impact of electrode material production. Furthermore, our extensive morphological investigations have demonstrated the impact of the nickel oxide content on the porous structure and electrochemical behavior of the electrode material. The properties of the PCNiO composites were characterized by X-ray diffraction (XRD), Fourier transform infrared (FTIR), field emission scanning electron microscopy (FESEM), Brunauer–Emmett–Teller (BET), thermo-gravimetric analysis (TGA), and Raman spectroscopy methods. The electrochemical properties of the PCNiO electrodes were tested using a two-electrode system. At a current density of 1.5 A/g , the GCD revealed that the C_{sp} of the PCNiO-150 electrode was 656.2 F/g . According to the CV curve, the C_{sp} range was 598.6 F/g at a scan rate of 10 mV/s . Additionally, even after 5000 cycles, the electrode had Coulombic efficiency of 97.55% and a strong cycle stability, with capacitance retention of 78.34%. The PCNiO-150 electrode also demonstrates a high ED of 44 Wh kg^{-1} and a PD of 562.5 W kg^{-1} . These results indicate that the integration of nickel oxide into activated porous carbon demonstrates significant improvements in the specific capacitance, cycle stability, and rate performance, which can be ascribed to its increased surface area and enhanced ion diffusion rates. Moreover, the combination of flexibility and low production costs indicates that the fabricated capacitors can fulfill the performance requirements for future mass production.

2. EXPERIMENTAL METHODS

2.1. Materials Used. The coconut shell was obtained from a local store, Mangalore, India. Nickel(II) oxide (NiO , trace metal 99.8%), potassium hydroxide (KOH), polyvinylidene

fluoride (PVDF), and charcoal were purchased from Sigma-Aldrich, India. The solvents employed were *N*-methyl-2-pyrrolidone (NMP), hydrochloric acid (1 M HCl), and distilled water (DW).

2.2. Preparation of Coconut-Shell-Derived Activated Porous Carbon. The raw coconut shells were washed several times in distilled water to eliminate impurities before being dried at 100 °C overnight. After drying, the shells were blended into a fine powder. Then, 10 g of coconut-shell powder was dispersed in 20 mL of 2 M KOH and heated to 100 °C for 12 h in an oven. After that, the resultant product was ground in a mortar and carbonized in a tubular furnace at 400 °C for 2 h under N₂ atmosphere. The final product was repeatedly filtered through 1 M HCl and distilled water and dried in the electric oven at 60 °C for 12 h. The resultant activated porous carbon is denoted as (PC).

2.3. Synthesis of an Activated Porous Carbon Nickel Oxide Nanocomposite. The PCNiO was synthesized using a simple hydrothermal process. Typically, 0.5 g of activated porous carbon (PC) was added to 50 mL of distilled water and stirred for 1 h. Then, 50 mg of NiO nanoparticles were added into the solution and stirred for 1 h. The resultant mixture was transferred to a Teflon-lined autoclave in a properly sealed metal jar of hydrothermal set up to treat heat for 12 h at 180 °C. After hydrothermal processing, the resulting solution was filtered and repeatedly washed with distilled water until the filtrate was clear. The produced solid mixture was dried at 60 °C in an air oven for 12 h. A similar repeated procedure is followed for different concentrations of nickel oxide nanoparticles (0, 50, 100, and 150 mg) coded as PCNiO-0, PCNiO-50, PCNiO-100, and PCNiO-150, respectively.

2.4. Fabrication of a Symmetry Supercapacitor Device. The electrochemical investigation was conducted by utilizing a two-electrode system. In this process, 3 g of KOH was dissolved in 30 mL of DW under vigorous stirring, and 3 g of PVA was dissolved in another 30 mL of DW also under vigorous stirring. The two separately prepared solutions were completely combined and then heated for 2 h at 80 °C while being constantly stirred to form a gel electrolyte.⁴⁸ All dissolved PCNiO nanocomposite (90%), PVDF (5%), and charcoal black (5%) are dissolved in a NMP solvent and completely mixed to form a slurry.⁴⁹ The resulting mixture is coated on a strip of aluminum (1 × 1 cm²) and dried for 10 h in a vacuum oven at 50 °C to obtain electrodes. The mass of the active material loading on each electrode was 1.2 mg cm⁻². For the fabrication of the symmetric supercapacitor (SSC), two working electrodes acted as both the positive and negative electrodes, utilizing PVA/KOH as the electrolyte for device assembly.⁵⁰

The specific capacitance and electrochemical efficiency of the electrode were evaluated using a variety of experimental techniques, including cyclic voltammetry (CV), galvanostatic charging–discharging (GCD), and electrochemical impedance spectroscopy (EIS). These investigations were conducted with the assistance of a CHI 660E electrochemical workstation model. The specific capacitance (C_{sp}) is obtained from the CV curve by using the following equation

$$C_{sp} = \frac{\int IdV}{q \cdot m \Delta V} (F g^{-1}) \quad (1)$$

The Galvano charging–discharging curves can be used to calculate the C_{sp} according to the following equations

$$C_{sp} = \frac{I \times \Delta t}{m \times \Delta V} (F g^{-1}) \quad (2)$$

where I (A) represents the current in both charging and discharging operations, while the quantity of the active electrode material is denoted as “ m ” (gm). The scan rate is denoted as q (V/s), the change in potential is represented by ΔV (volts), and the discharge time is denoted as “ Δt ” (s).

The Coulombic efficiency (η) of the supercapacitor was determined by using the subsequent equation

$$\eta = \frac{\Delta t_d}{\Delta t_c} \quad (3)$$

where Δt_d and Δt_c denote the time of discharge and charge, respectively.

The practical application of supercapacitors (SCs) is determined by two critical factors: the energy density (ED) and power density (PD), which can be calculated using the following equations

$$ED = \frac{C_{sp} \times \Delta V^2}{7.2} (Wh kg^{-1}) \quad (4)$$

$$PD = \frac{3600 \times E}{\Delta t} (W kg^{-1}) \quad (5)$$

2.5. Characterization Techniques. The structural change was examined using a benchtop X-ray diffractometer, Rigaku 600 miniflex (XRD). A Fourier transform infrared (FTIR) spectrometer (model α BRUKER) was used to analyze the chemical interaction. The surface morphology was examined using field emission scanning electron microscopy (Sigma Zeiss) and high-resolution transmission electron microscopy (model JEOL, JEM-2100). High spectroscopic resolution Raman images were provided by the Lab RAM HR (UV) spectroscopic equipment. Thermogravimetric and differential thermal analyses (TG/DTA) were conducted using the SDT Q600 V20.9 Build 20 instrument to investigate the thermal stability of the electrode materials. The surface area and pore-size distribution were examined using the Brunauer–Emmett–Teller (BET) model (Autosorb IQ-XR-XR, Anton Paar, Austria).

3. RESULTS AND DISCUSSION

3.1. X-ray Diffraction Analysis. XRD was used to examine the structural phase of the PCNiO nanocomposite, as shown in Figure 1. It is seen that there are two diffraction peaks at $2\theta = 23.5$ and 43° , corresponding to the (002) and (100) index planes of the conventional graphite structure for pure PCNiO-0.⁵¹ The presence of graphitization has suggested a major increase in the electroconductivity.³⁵ It is observed that the carbon peak (002) becomes broad and (100) clearly disappears with NiO nanoparticles. Moreover, the NiO diffraction peaks showed at $2\theta = 37, 43,$ and 62.5° for PCNiO-50 and PCNiO-100 nanocomposites, corresponding to the crystalline phase of NiO represented by the (111), (200), and (220) lattice planes.^{44,52} It is seen that three sharp and narrow peaks in PCNiO-150 indicate that a larger particle size could enhance the crystallinity. An additional diffraction peak observed at $2\theta = 75^\circ$ suggests the presence of a crystalline phase that corresponds to the (311) lattice plane.⁵³ In addition, the intensity of diffraction peaks was gradually increased with increasing concentration of NiO. The observed

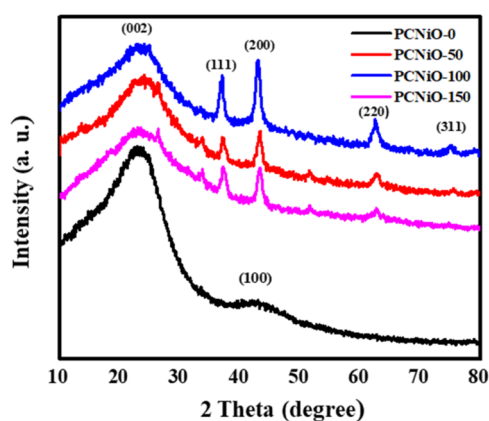


Figure 1. XRD pattern spectra of PCNiO-0, PCNiO-50, PCNiO-100, and PCNiO-150 nanocomposites.

result of XRD patterns clearly showed that porous carbon enhanced the porosity with increasing concentration of NiO, resulting in improved surface area of the material and is a suitable candidate for potential application in supercapacitors and other storage systems.

3.2. FTIR Analysis. FTIR was used to understand chemical functional groups in the synthesized activated porous carbon, and its PCNiO-0, PCNiO-50, PCNiO-100, and PCNiO-150 nanocomposites are depicted in Figure 2. The main peak of

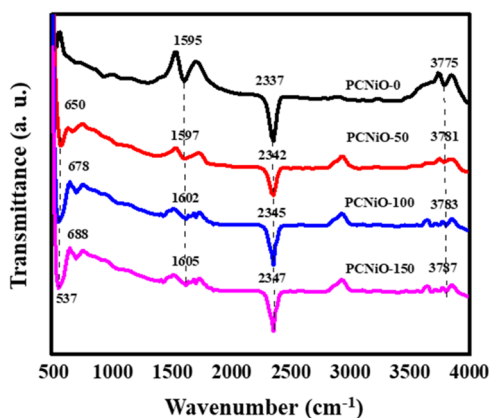


Figure 2. FTIR spectra of PCNiO-0, PCNiO-50, PCNiO-100, and PCNiO-150 nanocomposites.

pure carbon (PCNiO-0) is observed at 3775 cm^{-1} assigned to the stretching of the O–H bond⁵⁴ and a sharp band at 2337 cm^{-1} is due to the interaction of activated porous carbon and composite with CO_2 molecules.⁵⁵ The peak at 1595 cm^{-1} corresponds to the C=C stretching mode in the aromatic ring.⁵⁶ Furthermore, the incorporation of Ni ions demonstrated a significant change in the nanocomposite. Hence, the intensity of the O–H peaks observed in PCNiO-50, PCNiO-100, and PCNiO-150 displays a decrease with a slight deviation from the lower frequency. In addition, appearing of new bands at 650 and 537 cm^{-1} are attributed to the Ni–O stretching mode and Ni–O–H vibration, respectively.⁴³ Moreover, the intensities of vibration bands in the PCNiO-50, PCNiO-100, and PCNiO-150 composites decreased with the increase in the NiO concentration, which confirms the interaction of Ni ions with functional groups of activated porous carbon.

3.3. FESEM Analysis. FESEM images of PCNiO-0, PCNiO-50, PCNiO-100, and PCNiO-150 nanocomposites are depicted in Figure 3a–d, respectively. The pure PCNiO-0 showed a smooth surface and contains a thick wall with an irregular deep porous structure developed because of the chemical activation of the coconut shell, as shown in Figure 3a. The porous structure of PCNiO-0 can offer sufficient area for the deposition of nickel oxide. In addition, porous structures provide flexible network for charge diffusion through electrolytes.¹⁹ Figure 3b shows the FESEM image of PCNiO-50 that demonstrated a porous morphology with the loading of NiO nanoparticles on the surface of activated porous carbon (PC). However, the PCNiO-100 surface undergoes changes from smooth to rough with an irregular porous structure, as shown in Figure 3c.⁵⁷ Lastly, the PCNiO-150 shows roughness with a high porous structure due to nickel ion dispersion on the surface, as depicted in Figure 3d. The porous structure is regarded as highly advantageous for the electrode material in improving the specific capacity and rate capability as it acts like an ion-buffering storage to reduce the transport distance of ions to the inner surface of the electrode.⁵⁸

The energy dispersive X-ray spectra (EDS) were used to confirm the presence of NiO in the PCNiO nanocomposite, as shown in Figure 4a. The PCNiO-0 sample shows the presence of C and O. After incorporation of NiO nanoparticles onto the surface of PC with different concentrations of PCNiO-50, PCNiO-100, and PCNiO-150, the nanocomposites display the presence of C, O, and Ni, as shown in Figure 4b–d.⁴¹ Moreover, the concentration of Ni in PCNiO-150 is higher than those in PCNi-50 and PCNiO-100 while the concentrations of C and O are the lowest.

3.4. High-Resolution Transmission Electron Microscopy (HRTEM) Analysis. In order to further characterize the morphology of the PCNiO-0 and PCNiO-150 nanocomposite samples, TEM and HRTEM images were studied, are shown in Figure 5. The interconnected porous layer structure is clearly observed in the PCNiO-0 sample (Figure 5a,b). The HRTEM image in Figure 5c further supports the developed interconnected porous structure of PCNiO-0 and an apparent spacing of 0.34 nm associated with the (002) plane for graphite carbon.⁵⁹ The interconnected porous structure also functions as an electrolyte reservoir, which enables fast ion access into the porous carbon wall and hence the fast electron transfer and charge storage.⁶⁰ After loading NiO nanoparticles (NPs) on the surface of PC, the TEM image of the PCNiO-150 nanocomposite reveals a porous structure, and the NiO NPs are uniformly dispersed on the surface of the PC, as shown in Figure 5d,f. The disordered structure points to graphitic carbon, and the defects on the wall of the macropores ensure that there is the formation of mesopores due to the loading of NiO NPs on the PC surface. In the HRTEM image of the PCNiO-150 nanocomposite depicted in Figure 5e, the polycrystalline nature of the nickel species and lattice fringes with d -spacing of 0.21 and 0.241 nm were associated with the (200) and (111) crystallographic planes, which supported the feature of the cubic NiO structure, respectively,⁴² which is highly associated with the XRD results. Furthermore, the elemental mapping and composition of the PCNiO-150 nanocomposite were conducted by using STEM analysis. The NiO NPs are well-anchored on the surface of the porous carbon, and the C, O, and Ni elements are well-dispersed in the carbon layer, which is depicted by the element mapping as shown in Figure 5g.

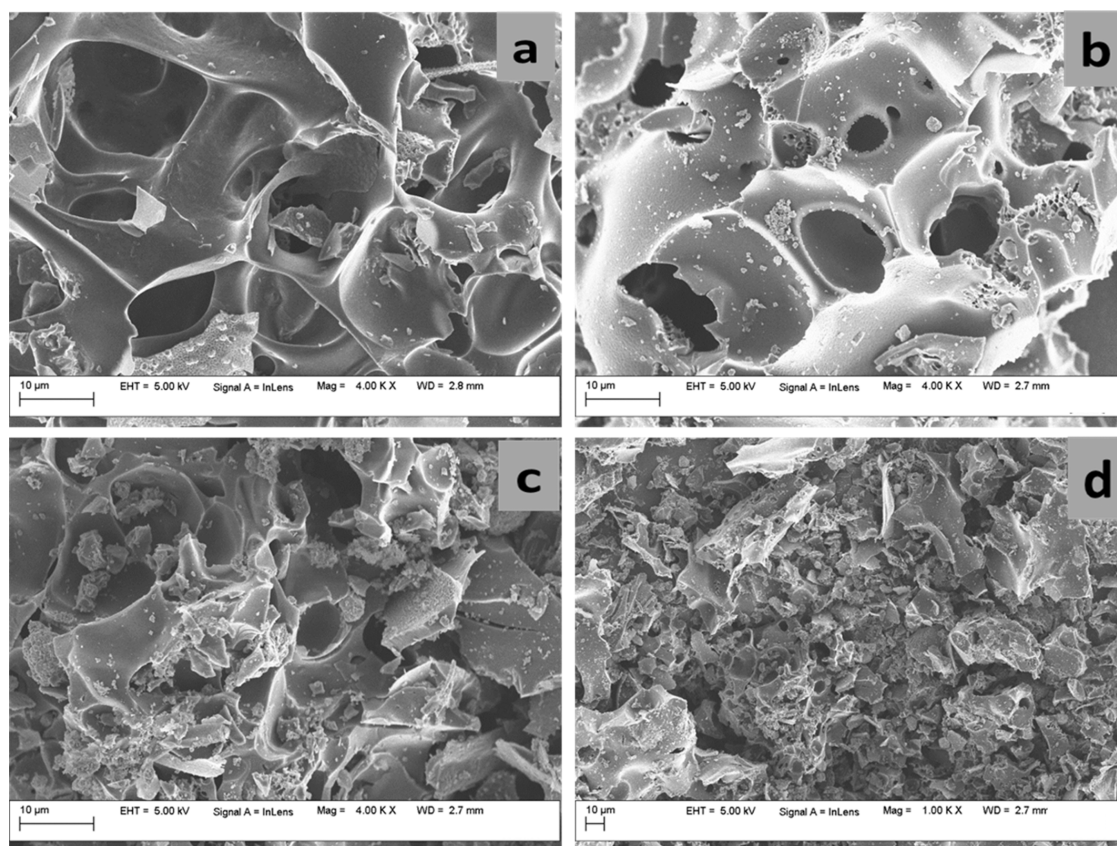


Figure 3. FESEM images of (a) PCNiO-0, (b) PCNiO-50, (c) PCNiO-100, and (d) PCNiO-150 nanocomposites.

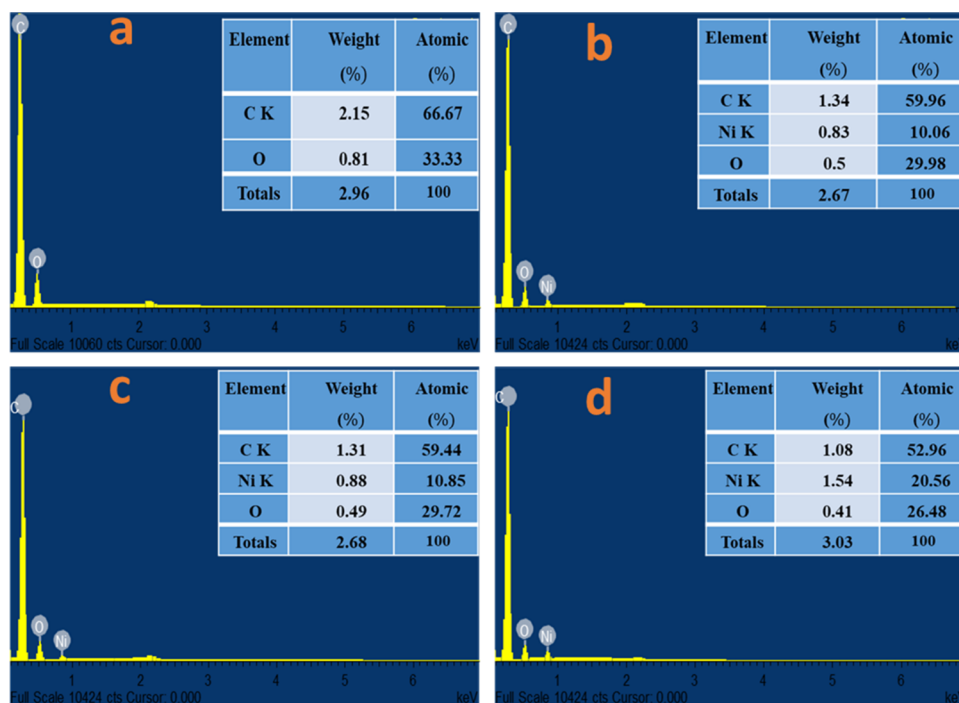


Figure 4. EDS images of (a) PCNiO-0, (b) PCNiO-50, (c) PCNiO-100, and (d) PCNiO-150 nanocomposites.

3.5. Raman Spectral Analysis. Raman spectroscopy is employed to determine the energy bands in the PCNiO nanocomposite, as illustrated in Figure 6. The Raman spectra of all samples (Figure 6) exhibit noticeable peaks at 1353 and 1592 cm^{-1} , which correspond to the distinct D and G bands,

respectively. The D-band signifies the mode (A_{1g}) caused by defects resulting from disordered carbon atoms, while the G-band indicates the mode (E_{2g}) originating from the ordered arrangement of carbon atoms in a graphitic structure.^{61,62} It is interpreted that the ratio of integrated intensities between the

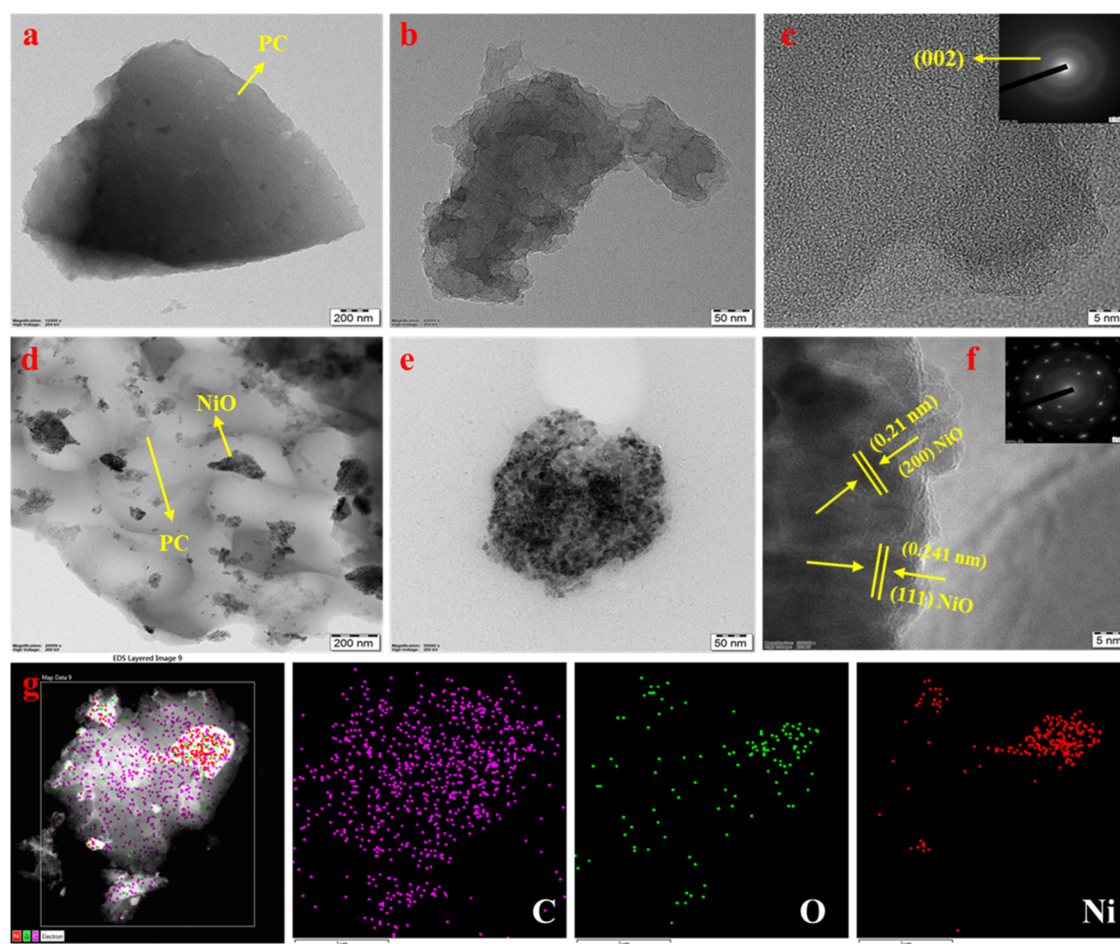


Figure 5. (a, b) TEM and (c) HRTEM images of PCNiO-0 (inset: corresponding selected-area electron diffraction (SAED) pattern), (d, e) TEM and (f) HRTEM images of PCNiO-150 (inset: corresponding SAED pattern), and (g) STEM elemental mapping analysis of PCNiO-150.

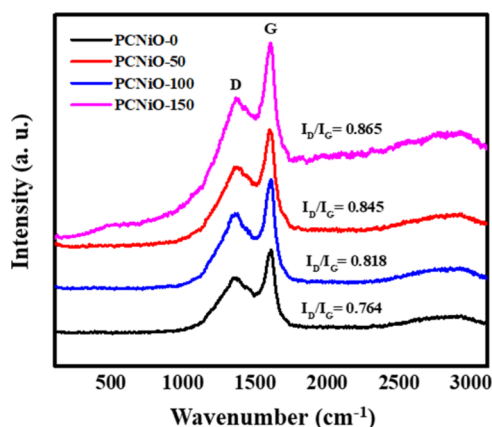


Figure 6. Raman spectra of PCNiO-0, PCNiO-50, PCNiO-100, and PCNiO-150 nanocomposites.

D and G bands (I_D/I_G) is indicative of the level of structural order in carbon materials concerning their carbonization. A reduced ratio corresponds to an increased degree of carbonization.⁶³ The I_D/I_G ratio for PCNiO-150, (0.865), PCNiO-100, (0.845), and PCNiO-50, (0.818) is observed to be higher than that of PCNiO-0 (0.764), as depicted in Figure 6. An increase in the I_D/I_G intensity ratio suggests a reduction in the dimensions of the in-plane sp^2 domains and a structurally organized crystalline pattern of PCNiO-50, PCNiO-100, and

PCNiO-150 composites,⁶⁴ indicating the occurrence of more porosity as a result of improvement of the surface area due to NiO nanoparticle, leading to the formation of storage sites and also easy charge diffusion.

3.6. BET Analysis. The nitrogen adsorption–desorption isotherm was employed for determining the specific surface area, pore dimensions, and the distribution of pore sizes in the PCNiO-0, PCNiO-50, PCNiO-100, and PCNiO-150 nanocomposites. As depicted in Figure 7a, the observed results indicated that both (pure) porous carbon and the nanocomposite display type IV isotherms according to IUPAC classification. Notably, a distinct hysteresis loop is observed at relative pressures of (1–0.2) for PCNiO-0, (1–0.5) for PCNiO-50, and (1–0.4) for PCNiO-150, and it demonstrates the adsorption characteristics of mesoporous substances.^{65,66}

Specifically, for PCNiO-0, it was observed that the specific surface area is $21.16 \text{ m}^2 \text{ g}^{-1}$, and the average pore size is 2.4 nm. This result shows a great deal of mesopores for the PCNiO-0 sample, which suggests that activated carbon has the potential to serve as the main support structure for bonding with nickel oxide. In contrast, PCNiO-50 and PCNiO-150 have higher specific surface areas of 39.60 and $30.63 \text{ m}^2 \text{ g}^{-1}$. These nanocomposites also observed average pore sizes centered around 3.18 and 2.68 nm, as shown in Figure 7b. Furthermore, PCNiO-150 showed a relatively lower pore volume and smaller specific surface area compared to PCNiO-50, as shown in Table 1. This clearly indicates that the

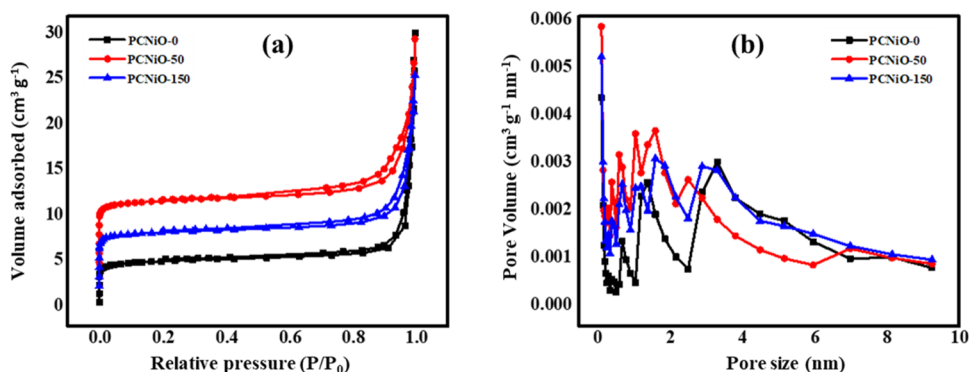


Figure 7. (a) Nitrogen adsorption/desorption isotherms and (b) pore-size distribution curves of PCNiO-0, PCNiO-50, and PCNiO-150 nanocomposites.

Table 1. Characteristics of the Pore Structures of PCNiO-0, PCNiO-50, and PCNiO-150 nanocomposites

samples	specific surface area (m² g⁻¹)	pore volume (cm³ g⁻¹)	average mesopore diameter (nm)
PCNiO-0	21.16	0.013	2.40
PCNiO-50	39.60	0.031	3.18
PCNiO-150	30.63	0.02	2.68

incorporation of more NiO fills the mesopores within the activated carbon. As a result, the mesoporous structure in the composites characterized by a substantial pore volume enhances active sites for electrode–electrolyte interactions and reduces ion diffusion distances.⁶¹ In addition, the integration of NiO significantly improves the overall efficacy, resulting in an increased surface area that enhances surface wettability and promotes pseudocapacitive attributes.

3.7. TGA Characterization. The TGA and DTA were used to analyze the thermal stabilities of PCNiO-0, PCNiO-50,

PCNiO-100, and PCNiO-150 nanocomposites from room temperature to 900 °C with a heating rate of 10 °C min⁻¹, as depicted in Figure 8a–d. The TGA curve of the PCNiO-0 sample demonstrated three main regions: the evaporation of water molecules (up to 150 °C with 11.6% weight loss),⁶⁷ degradation of residual organic materials in coconut fibers (150–550 °C with 67.6% weight loss), and completion of the activated carbonization process (550–900 °C with 11.2% weight loss)³² as shown in Figure 8a. It is seen that the two main regions of weight loss curves of PCNiO-50, PCNiO-100, and PCNiO-150 nanocomposites are shown in Figure 8b–d. The first region is due to the desorption of surface-bound water up to 250 °C, and the second region is due to the decomposition of oxygen-containing groups and the carbon substrate, which occurs between 250 and 650 °C. Additionally, the significant reduction in mass starts at around 452 °C for PCNiO-0 as it was detected in the TDA curve. Furthermore, the PCNiO-50 is recorded, and this peak at approximately 461 °C is sharp. On the other hand, PCNiO-100 and PCNiO-150

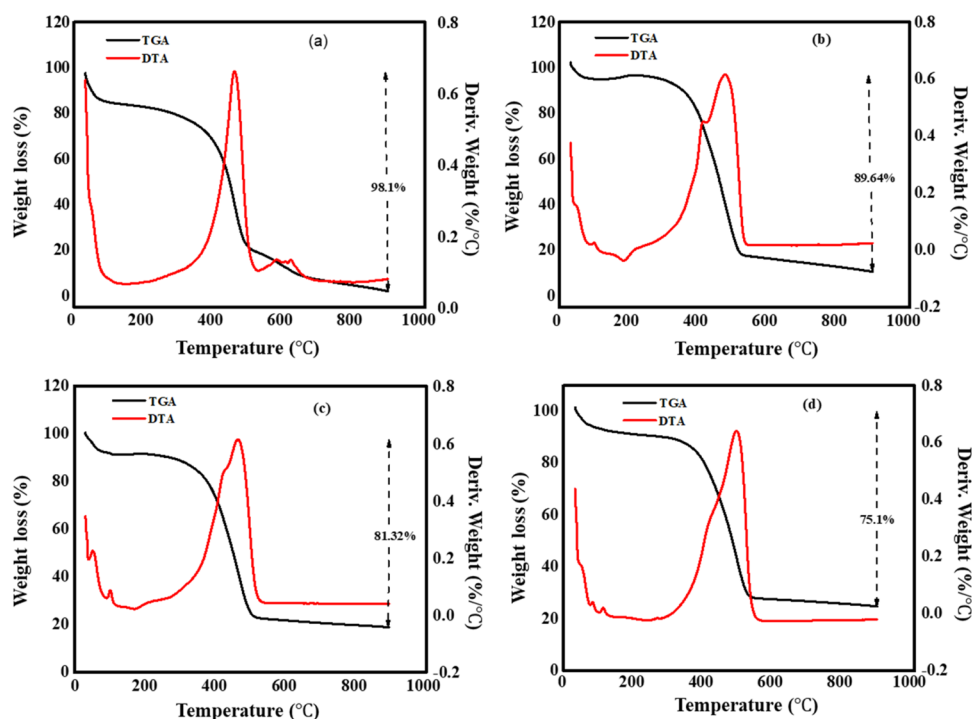


Figure 8. TGA/DTA curves of (a) PCNiO-0, (b) PCNiO-50, (c) PCNiO-100, and (d) PCNiO-150 nanocomposites.

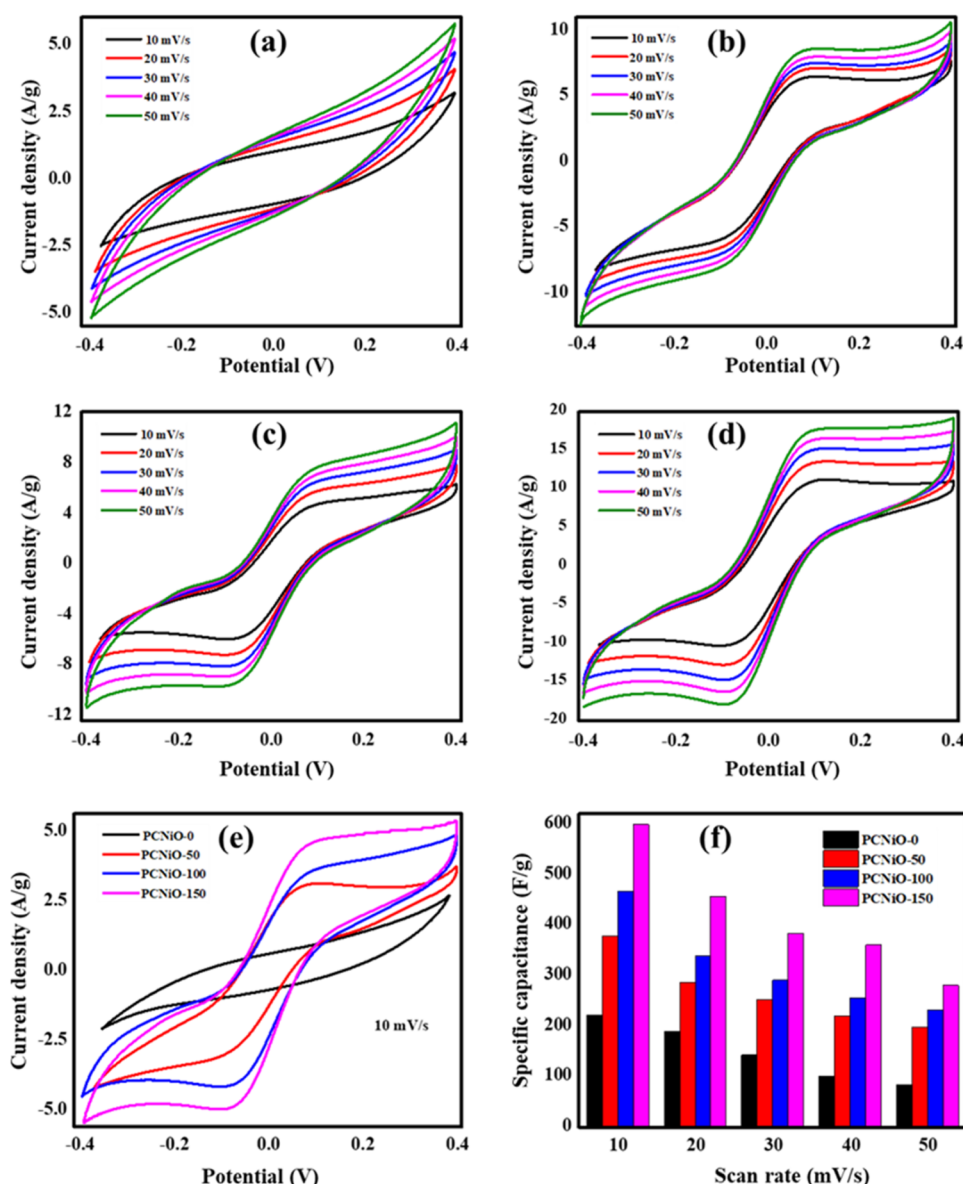


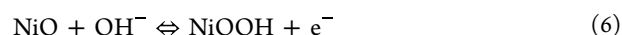
Figure 9. CV curves of (a) PCNiO-0, (b) PCNiO-50, (c) PCNiO-100, (d) PCNiO-150, and (e) CV curves of PCNiO-0, PCNiO-50, PCNiO-100, and PCNiO-150 at a scan rate of 10 mV/s. (f) Variation of specific capacitance for scan rates from 10 to 50 mV/s.

showed a shift in this characteristic peak, with the peak appearing at 468 and 489 °C, respectively. These results confirm the involvement of NiO molecules in the intermolecular structure of the PC specimen, influencing both its thermal properties and physical characterization. Furthermore, the total weight loss for PCNiO-50, PCNiO-100, and PCNiO-150 nanocomposites were 89.64, 81.32, and 75.1 wt %, respectively, and they agree well with the expected NiO loading.

3.8. Electrochemical Performance of a PCNiO Symmetric Supercapacitor. **3.8.1. Cyclic Voltammetry.** The electrochemical behavior of the synthesized electrodes was carried out in the two-electrode system by using cyclic voltammetry analysis. Figure 9 demonstrates the CV curves of PCNiO-0, PCNiO-50, PCNiO-100, and PCNiO-150 with a potential window ranging from −0.4 to +0.4 V at scan rates from 10 to 50 mV/s.

The CV curve of PCNiO-0 exhibits characteristics of both double-layer behavior and redox behavior. This may result

from surface contaminants and functional groups introduced during the activation method with KOH.²⁶ While the CV curves of all composite electrodes typically exhibit a couple of redox peaks with a symmetric shape, this indicates pseudocapacitive behavior due to the presence of NiO and double-layer capacity behavior of the activated porous carbon.^{42,44} The anodic peak at 0.1 V results from the oxidation of NiO to NiOOH, whereas the cathodic peak at −0.12 V indicates the reverse process, as illustrated in the redox reaction given by the following equation



The specific capacitance (C_{sp}) values of PCNiO electrodes were calculated using eq 1 and an estimated value of 221.5, 378.7, 467.2, and 598.6 F/g at a scan rate of 10 mV/s for PCNiO-0, PCNiO-50, PCNiO-100, and PCNiO-150 electrodes, respectively, as depicted in Figure 9a–d. The PCNiO-150 electrode typically shows an increase in the area of the CV curve compared with other PCNiO-0, PCNiO-50, and

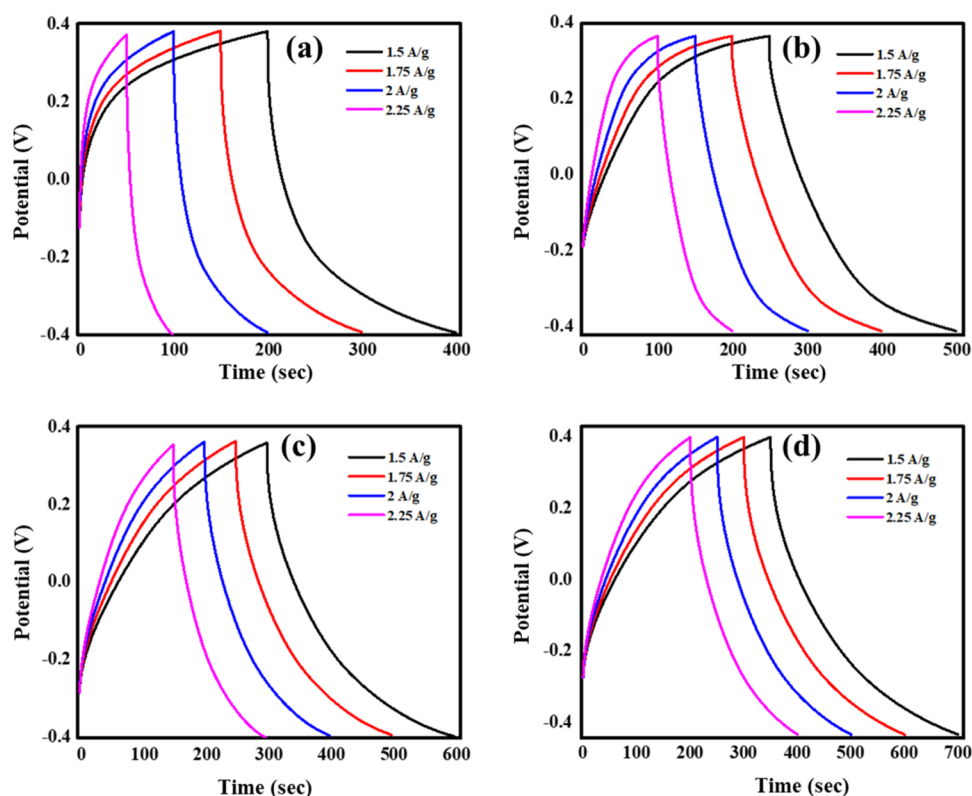


Figure 10. GCD curves of (a) PCNiO-0, (b) PCNiO-50, (c) PCNiO-100, and (d) PCNiO-150.

PCNiO-100 electrodes, which clearly indicates the increasing capacitance of the PCNiO-150 electrode as observed in Figure 9e. This indicates that the porous carbon in the PCNiO-150 nanocomposite offers high conductivity and a porous network with an extensive surface area, hence enabling fast electrolyte-ion transport, which enhances its specific capacitance. PCNiO-100, despite having a porous structure, has a smaller area compared to PCNiO-150, which has a highly porous nature. In contrast, PCNiO-0 has a smaller area due to its poor morphological features. Furthermore, the capacitance decreases when the scan rate increases, as observed by the comparison bar chart in Figure 9f.⁵⁰ This behavior is due to the limitation of electrolyte ion diffusion during the redox process to satisfy electronic neutralization at the electrode–electrolyte interface.⁴³ Additionally, at higher scan rates, OH^- ions can reach the outer surface of the electrode. Conversely, at lower scan rates, OH^- ions effectively intercalate with both the outer and inner pore surfaces. The limited access of OH^- ions leads to a decrease in the specific capacitance at higher scan speeds.

3.8.2. Galvanostatic Charging–Discharging Analysis. The GCD curves of the PCNiO-0, PCNiO-50, PCNiO-100, and PCNiO-150 electrodes were measured at a potential range of -0.4 to 0.4 V and at various current densities from 1.5 to 2.25 A/g, as shown in Figure 10. GCD curves for all electrodes typically show nonlinear and symmetric charging–discharging shapes, which are characteristic of the capacitive behavior of activated porous carbon and redox reaction of NiO.⁶⁸ The specific capacitance (C_{sp}) was calculated from the GCD curves by using eq 1. The PCNiO-150 electrode demonstrated a C_{sp} value of 656.2 F/g at a CD of 1.5 A/g, which is higher than for PCNiO-100 (562.5 F/g), PCNiO-50 (468.7 F/g), and PCNiO-0 (375 F/g), as depicted in Figure 10a–d and listed in Table 2. The C_{sp} of the PCNiO nanocomposite electrodes

Table 2. Specific Capacitances of PCNiO-0, PCNiO-50, PCNiO-100, and PCNiO-150 Calculated by the GCD Test at Different Current Densities

current density (A/g)	specific capacitance (F/g)			
	PCNiO-0	PCNiO-50	PCNiO-100	PCNiO-150
1.5	375	468.7	562.5	656.2
1.75	334.3	407.8	502.4	594.8
2	258.2	346	432.1	503.4
2.25	187.5	234.3	327.6	417.5

decreases with increasing current density, as observed by the bar chart in Figure 11a. The decrease in capacitance is due to an insufficient reaction with a higher current density, leading to inevitable consequences. At low current densities, ions intercalate and deintercalate due to sufficient time, allowing access to both the outer surface and inner pores. But at a high current density, ions only embed on electrode material surfaces, causing specific capacitance to decline somewhat.

As a result of the GCD and CV studies, the C_{sp} values of PCNiO electrodes from cyclic voltammetry are lower than the charge–discharge results. Furthermore, the PCNiO-150 electrode has a higher capacitance than other electrodes, attributed to the greater number of defects, abundant porous carbon network, robust microstructures, and large surface area of the composite framework. This can facilitate the transport of ions in the material, reducing the ionic resistance and improving the charging–discharging kinetics.⁵⁸

Electrochemical impedance spectroscopy (EIS) is a powerful technique used to test the internal resistance and charge transfer resistance of the SSCs device. The Nyquist plot spectra for the PCNiO-0, PCNiO-50, PCNiO-100, and PCNiO-150 electrodes in frequencies ranging from 1 Hz to 100 kHz are

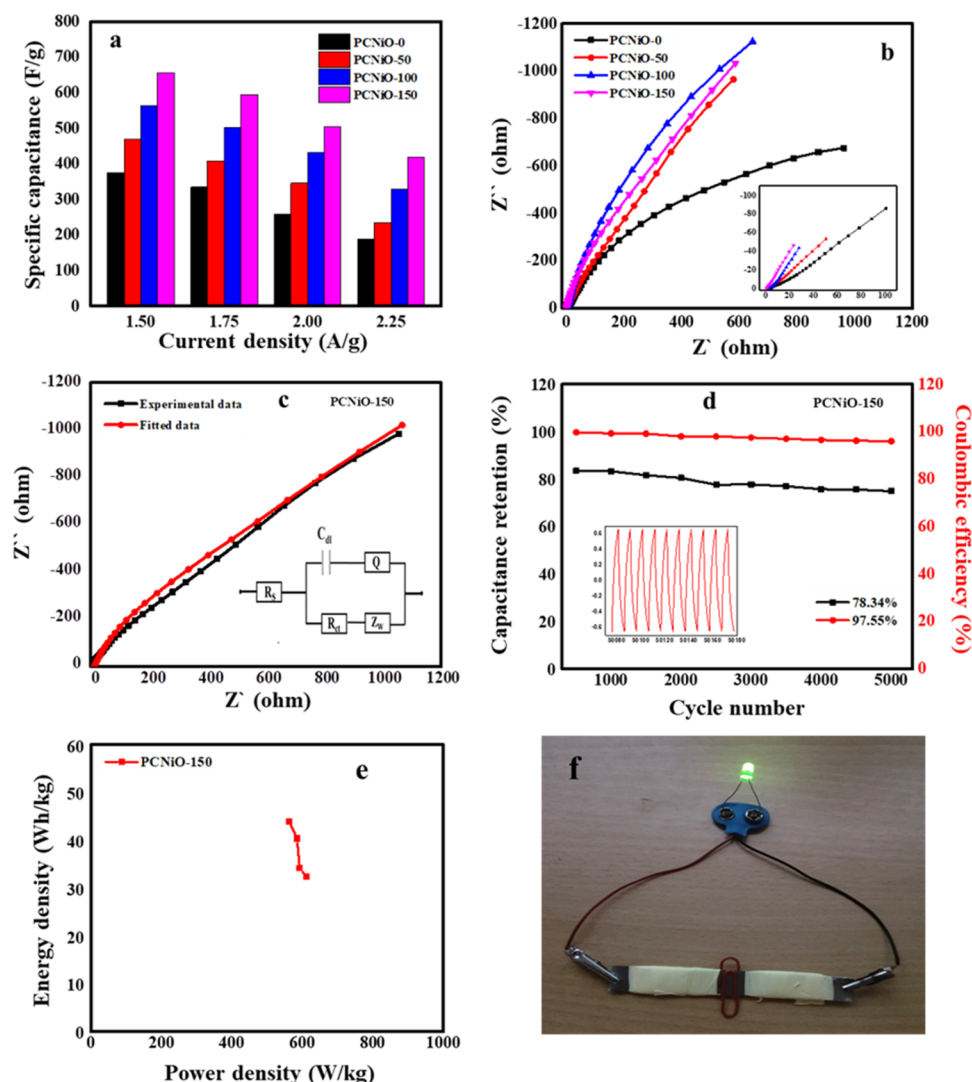


Figure 11. (a) Variation of C_{sp} for different CDs from 1.5 to 2.25 A/g, (b) Nyquist plots of PCNiO-0, PCNiO-50, PCNiO-100, PCNiO-150, (c) Fitted curve with equivalent circuit of PCNiO-150, (d) Capacitance retention and Coulombic efficiency of the PCNiO-150 SSC device; inset: the last 10 cycles of the GCD curve, (e) Ragone plots of the PCNiO-150 SSC device, and (f) digital image of the assembly PCNiO-150 SSC device glows with green LED.

depicted in Figure 11b. The plot is divided into two regions: (i) the high-frequency region demonstrates a semicircle and (ii) the low-frequency region demonstrates an inclined straight line. The semicircle indicates the charge transfer resistance (R_{ct}), which is related to the kinetics of the faradaic reactions that occur at the electrode–electrolyte interface, while the inclined line represents the Warburg impedance (Z_w), which is related to the diffusion of electrolyte ions into the porous electrode.⁶⁷ The inset of Figure 11b demonstrates that the diameter of the semicircle slightly decreases as the NiO concentration increases. The PCNiO-150 electrode shows lower charge transfer resistance, exhibits faster charge transfer kinetics, and can deliver energy more quickly. Additionally, the solution resistance (R_s) was determined from the line intercept with the Z' -axis at a high-frequency region.³² The solution resistance (R_s) values of PCNiO-0, PCNiO-50, PCNiO-100, and PCNiO-150 electrodes are 2.26, 1.33, 1.2, and 0.64 Ω , respectively.

Moreover, the PCNiO-150 nanocomposite electrode demonstrates a low R_s value, indicating that the electrolyte ions can diffuse more easily into the electrode, which suggests that this

electrode has a high storage capacity. The inset of Figure 11c also displays the Nyquist plot of the equivalent circuit for the PCNiO-150 electrode. The R_{ct} and the constant phase element (Q) are connected in series with R_s . The Warburg impedance (Z_w) is in series with the R_{ct} , which represents the diffusion from the high- to the low-frequency region. Typically, an ideal electrode with substantial capacitance C at very low frequencies is anticipated to generate a linear representation aligned with the $-Z''$ -axis.⁶⁹ However, a slight deviation from the optimal performance is observed, which may be attributed to the resistive component linked to C , which is the leakage resistance.

The reliability of a supercapacitor is an important factor to consider in terms of its long-term performance and suitability for various applications. As a result, the PCNiO-150 SSC device exhibits excellent cycling stability even after 5000 charge–discharge cycles. At a current density of 1.5 A/g, the PCNiO-150 SSC device retained 78.34% of its initial capacitance after 5000 cycles. The Coulombic efficiency is expressed as a percentage and calculated using eq 3. The PCNiO-150 SSC device exhibits a high Coulombic efficiency

of 97.55% after 5000 charge–discharge cycles, as shown in Figure 11d. A high Coulombic efficiency indicates that the PCNiO-150 SSC device can efficiently store and deliver charge without significant losses.

The energy density (ED) and power density (PD) are important parameters for evaluating the SSC device performance; they have been calculated using eqs 4 and 5, respectively. The PCNiO-150 SSC device demonstrates a high ED of 44 Wh kg^{−1} at a PD of 562.5 W kg^{−1}, and it decreases to 28.4 Wh kg^{−1} at a maximum PD of 611.4 W kg^{−1}, as depicted by the Ragone plot in Figure 11e. The values of ED and PD for the PCNiO-150 electrode are slightly higher than other porous carbon/NiO nanocomposite electrodes for SCs described recently, such as NiO//AC (52.4 Wh kg^{−1} at 800 W kg^{−1}),⁵² NiCo LDH/IPC//IPC (29.6 Wh kg^{−1} at 744 W kg^{−1}),²² C/NiSi-600-1 (25.24 Wh kg^{−1} at 551.4 W kg^{−1}),⁶⁸ 3D NC/Ni@NiO (19.4 Wh kg^{−1} at 700 W kg^{−1}), and⁷⁰ Ni/N-doped PC (41.14 Wh kg^{−1} at 800 W kg^{−1}),⁷¹ as depicted in Table 3.

Table 3. Comparison of the Results with Recent Work on Porous Carbon and Nickel Oxide Nanocomposites

samples	CD (A/g)	C _{sp} (F/g)	ED (Wh kg ^{−1})	PD (W kg ^{−1})	references
3D NC/Ni@NiO	1	493	19.4	700	70
NiO//AC	0.5	568.7	52.4	800	52
Ni/N-doped PC	1	2002.6	41.14	800	71
NiCo LDH/IPC//IPC	-	-	29.6	744	22
C/NiSi-600-1	0.5	237.07	25.24	551.4	68
PCNiO-150	1.5	656.2	44	562.5	this work

As illustrated in Figure 11f, two parts of the manufactured SSCs were employed for controlling a commercial green LED connected in series and its illumination for approximately 30–40 s during the experiment to demonstrate the efficient use of the application.

The high electrochemical performance of the PCNiO nanocomposite electrode could be attributed to several factors: (i) the synergistic effect of the pseudocapacitive contribution originating from NiO nanoparticles and the EDLC performance derived from the porous carbon network.⁷² Also, the carbon matrix enables efficient charge accommodation, high cycle durability, and rate capability. (ii) The PC offers a conducive grid and structural strength, whereas nickel oxide boosts the particular capacitance via its pseudoelectrical contribution. This boosts transmission and diffusion for higher capacitance.⁷³ (iii) The rich mesoporous structure provided ion-buffering reservoirs, which enhanced the wettability and ion exchange of the electrolyte by materials, increased the active sites for ion adsorption, and improved the electrochemical performance of the materials. These outcomes provide an affirmation of the applicability of the PCNiO nanocomposite as the electrode material for the development of a supercapacitor.

4. CONCLUSIONS

In conclusion, the incorporation of nickel oxide into activated porous carbon derived from coconut shell (PCNiO) is a promising approach to develop new composite materials with enhanced properties for supercapacitor applications. The PCNiO nanocomposite was characterized and analyzed using

a variety of techniques, such as XRD, FTIR, FESEM, HRTEM, BET, TGA, and Raman spectroscopy. These techniques revealed the presence of a mesoporous structure with a flexible network that facilitates the rapid diffusion of ions, and the synergistic effect between NiO and porous carbon contributes to boosting the electrochemical performance. The electrochemical properties of PCNiO samples are carried out in the PVA/KOH electrolyte by using a two-electrode system. Among the prepared samples, the PCNiO-150 electrode demonstrates a maximum specific capacitance of 656.2 F/g at a rate of 1.5 A/g. The cyclic voltammetry curve of PCNiO-150 shows a high capacitance of 598.6 F/g at a scan rate of 10 mV/s. Even after 5000 cycles, it had extreme cycle stability, with capacitance retention of 78.34% and Coulombic efficiency of 97.55%. Additionally, the PCNiO-150 SSC shows a high energy density of 44 Wh kg^{−1} and a power density of 562.5 W kg^{−1}. Additionally, the fabricated SSC device is serially connected to turn on a commercial green LED for 30–40 s at the time of the experiment. These results demonstrate that the incorporation of nickel oxide into activated porous carbon can significantly improve the specific capacitance, cycle stability, and rate performance in supercapacitors. Moreover, the use of low-cost and environmentally friendly synthesis methods makes these materials practical applications for large-scale production.

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Notes

The authors declare no competing financial interest.

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